STATEMENT OF WORK (SOW)

GrowPet Platform — Smart Care for Happy Pets

Project Type: Multi-species pet wellness and care platform

Technology: React 18 • TypeScript • Vite • Tailwind CSS • Firebase

Production Status: All seven planned milestones completed

1. Executive Summary

GrowPet is a multi-species digital wellness platform designed to help pet owners organize everyday care, health information, veterinary support, product discovery, and personalized routines in one place. The platform is intended to support mammals, birds, fish, reptiles, amphibians, and exotic pets rather than limiting care workflows to conventional household animals.

The solution combines species-aware user experiences with cloud synchronization, AI-assisted guidance, location-based veterinary discovery, medication and health records, routine tracking, and shopping-price comparison. The objective is to provide a practical digital companion for responsible pet ownership while keeping the interface simple enough for frequent daily use.

2. Project Objectives

- Create a unified care platform supporting multiple pet species and care requirements.
- Generate personalized routines and checklists based on the selected pet species.
- Provide AI-assisted, species-aware educational and triage guidance.
- Help users find nearby veterinary services using location-aware search.
- Maintain structured medication, health, weight, and vital records.
- Enable users to discover and compare pet-product prices across major retailers.
- Synchronize user data across browsers through a cloud-backed account.
- Deliver a responsive, production-ready web application.

3. Scope of Work

3.1 Universal Pet Profiles

The profile system provides a centralized identity for each pet and is designed to accommodate different species. Users can identify pets through smart species photo auto-matching, choose from a preset gallery, or upload an image from their device. Profiles act as the foundation for personalized routines, AI guidance, and health information.

Key capabilities:

- Smart species photo auto-matching
- Preset species gallery
- Device file/image upload
- Species-linked personalization
- Profile information used across care modules

3.2 Care Routine Engine

The Care Routine Engine transforms species-specific care requirements into actionable daily checklists. It is designed to encourage consistency through completion tracking, streaks, and a pet happiness score. This module turns general pet-care knowledge into a repeatable workflow.

Key capabilities:

- Auto-generated species-based care checklists
- Daily task completion tracking
- Streak counter
- Happiness score
- Routine-oriented dashboard experience

3.3 AI Vet Assistant

The AI Vet Assistant provides educational and decision-support guidance tailored to the selected species. Its intended use includes triage-oriented questions, nutrition information, behavioral guidance, and emergency-oriented recommendations. The feature is positioned as an informational assistant and should not be treated as a replacement for a licensed veterinarian.

Key capabilities:

- Species-aware responses
- Symptom/triage guidance
- Nutrition assistance
- Behavioral guidance
- Emergency guidance and escalation recommendations

3.4 GPS Vet Locator

The Vet Locator helps users identify veterinary services using location-aware search. The design supports device GPS as the primary location method with IP-based fallback and city-based search when appropriate. Users can open directions through Google Maps.

Key capabilities:

- Device GPS location
- IP fallback
- City/locality search
- Nearby veterinary discovery
- Direct Google Maps directions

3.5 Medication & Health Vault

The Health Vault provides a structured place for maintaining pet-care records. Medication events can be logged with timestamps, while weight and vital measurements can be retained as a historical record. This supports continuity of care and gives owners a clearer view of changes over time.

Key capabilities:

- Medication dose logging
- Timestamped health events
- Weight history
- Vital history
- Centralized pet health records

3.6 Price Comparison & Budget Tracking

The Price Comparison module is designed to help owners compare pet-product prices and manage spending. The stated retailer coverage includes Amazon, Chewy, Petco, PetSmart, and Walmart. The feature supports product discovery and budget awareness across multiple purchasing sources.

Key capabilities:

- Multi-retailer price comparison
  - Amazon, Chewy, Petco, PetSmart, and Walmart coverage
  - Budget tracking
  - Product-oriented purchasing workflow

3.7 Passwordless Cloud Sync

The cloud synchronization layer is designed to maintain user data across browsers using Firebase authentication and Cloud Firestore. A deterministic user identifier associates cloud records with the correct account, allowing supported data to remain consistent between browser sessions.

Key capabilities:

- Passwordless authentication flow
  - Cloud-backed Firestore storage
  - Deterministic UID association
  - Multi-browser synchronization
  - Real-time data synchronization where supported

4. Technology Stack

Layer	Technology	Purpose
Frontend	React 18	Component-based application interface
Language	TypeScript	Typed application development
Build Tool	Vite	Fast development and production builds
UI / Styling	Tailwind CSS	Responsive and utility-based styling
Authentication	Firebase Auth	User authentication and identity
Database / Sync	Cloud Firestore	Cloud data storage and synchronization
Hosting	Firebase Hosting	Production web deployment

5. Functional Architecture

At a high level, GrowPet follows a client-cloud architecture. The React/TypeScript frontend provides the user experience and application logic. Firebase Authentication manages identity, Cloud Firestore provides cloud persistence and synchronization, and Firebase Hosting serves the production web application. Location services support veterinary discovery and map navigation, while external retail sources support product-price comparison.

Primary flow:

User → Authentication → Pet Profile → Personalized Care Modules → Cloud Data / Sync → Supporting Services (AI, location, maps, retail sources)

6. User Experience & Interface Requirements

The platform is intended to provide a clear, responsive, and task-oriented interface. Core actions such as viewing a pet, completing care tasks, recording health information, finding a vet, asking the AI assistant, and checking product prices should be accessible without unnecessary navigation.

UX priorities include responsive layouts, consistent visual hierarchy, readable information presentation, clear completion states, straightforward navigation, and mobile-friendly interaction patterns.

7. Security & Data Considerations

Authentication and cloud data access are handled through Firebase services. User records should be associated with authenticated identities, and Firestore security rules should restrict access so that users can read and modify only authorized records. Sensitive health information should be treated as private user data, with appropriate access controls and minimal collection of unnecessary information.

The AI assistant should clearly communicate that its output is informational and that urgent or uncertain medical situations require professional veterinary care.

8. Milestones & Deliverables

Milestone	Deliverable	Status
1	UI/UX design and core application interface	Completed
2	Vet Locator and location-based discovery	Completed
3	Price Aggregator / comparison functionality	Completed
4	Authentication implementation	Completed
5	Cloud synchronization and Firestore integration	Completed
6	Media / image and profile functionality	Completed
7	Production deployment	Completed

9. Production Deployment

The GrowPet application has been deployed as a production web application through Firebase Hosting. The production URL supplied for the project is:

https://edunova-b432b.web.app

Note: The supplied production URL uses the edunova-b432b.web.app Firebase Hosting domain. This SOW records the URL exactly as provided; the deployment/domain name should be verified if the production site is intended to be branded publicly as GrowPet.

10. Testing & Acceptance

Acceptance is based on successful implementation and operation of the seven stated milestones. Testing should cover authentication, profile creation and media selection, routine generation/completion, AI assistant interactions, vet search and map navigation, health/medication logging, price comparison flows, cross-browser synchronization, responsive layouts, and production deployment.

11. Deliverables Summary

The completed GrowPet scope consists of a responsive multi-species pet wellness platform with universal profiles, species-aware care routines, AI-assisted guidance, veterinary discovery, health and medication records, product-price comparison, cloud authentication/synchronization, media support, and production hosting.

12. Future Enhancement Opportunities

Potential future releases could expand the platform with appointment booking, vaccination reminders, document/PDF storage, wearable or IoT integrations, richer analytics, veterinary-provider accounts, push notifications, offline-first support, multilingual experiences, advanced AI personalization, and native Android/iOS applications.

13. Project Completion Statement

Based on the project information supplied, all seven planned milestones have been marked as completed, including UI/UX, veterinary locator, price aggregation, authentication, synchronization, media functionality, and production deployment. GrowPet therefore represents a completed production-oriented project scope suitable for demonstration, portfolio presentation, and further iteration.

— End of Statement of Work —